CINEMATCH: SYSTEM PIPELINE & ARCHITECTURE

Complete Technical Workflow: Ingestion -> NLP Vectorization -> Reactive UI -> Cloudflare Tunnel

1. DATA INGESTION & FEATURE PIPELINE

DATABASE

Movie Metadata Matrix

> 30 Curated Blockbusters (Pandas DataFrame)

> TMDB CDN Verified Poster Assets

Merge

RATINGS

User Ratings Engine

> 140+ Explicit Interaction Records

> Sparse User-to-Item Matrix Format

Tokenize

PREPROCESS

Feature Synthesis

> N-gram Token Concatenation (Plot + Mood + Genre)

> Stopword Stripping & Corpus Normalization

Corpus Stream

2. NLP VECTORIZATION & SIMILARITY ENGINE

NLP MODEL

TF-IDF Vector Space

> Scikit-Learn TfidfVectorizer Sparse Matrix

> Term Frequency Inverse Document Weighting

Rank

SIMILARITY

Cosine Similarity Engine

> Mathematical Metric: (A . B) / (||A|| x ||B||)

> 30x30 Pairwise Score Matrix (0.0 to 1.0)

Compute

HYBRID

Collaborative & Hybrid Logic

> User-User Pearson / Cosine Correlation

> Combined Content & Collaborative Ranking

Ranked Results

3. STREAMLIT APPLICATION CORE (FULL STACK)

FRONTEND

Catalog & Smart Search

> Levenshtein Fuzzy Title & Actor Matching

> Streamlit @st.cache_data Memory Optimization

Query

EXPERIENCE

Interactive Systems

> Movie Roulette (Real-time Randomizer)

> 30-Sec 3-Question Decision Quiz Algorithm

Render

MOBILE

Mobile Responsive UI

> Custom CSS Mobile-First Media Queries

> Touch Targets & YouTube Trailer Deep-links

Build

4. EDGE HOSTING & MOBILE DISTRIBUTION

HOST

Localhost Server

> Python Runtime / Streamlit Server on Port 8501

> Desktop 1-Click Launcher (Start CineMatch.bat)

Tunnel Port

TUNNEL

Cloudflare Quick Tunnel

> Zero-Password Edge HTTPS (trycloudflare.com)

> No Port-Forwarding or Accounts Required

Broadcast

DISTRIBUTE

Instant Mobile Access

> Universal iOS Safari & Android Chrome Support

> 1-Click WhatsApp Direct Invite Generator